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Abundance and transmission of antibiotic resistance and virulence genes through mobile genetic elements in integrated chicken and fish farming system

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Integrated chicken and fish farming systems, common in Bangladesh, present significant public health risks due to the spread of antimicrobial resistance genes (ARGs) and virulence factors (VFGs) through mobile genetic elements (MGEs). This study employs metagenomic sequencing to explore the diversity and abundance of ARGs, VFGs, and MGEs in various environmental samples from these farming systems. A total of 384 ARGs were detected, with tetracycline resistance genes such as *tetM* and *tetX* being the most abundant, alongside macrolide-lincosamide-streptogramin and aminoglycoside resistance genes. Droppings harbored the highest proportion of ARGs (62.2%), whereas sediment served as a reservoir for multi-metal resistance genes. Virulence factors associated with immune modulation, such as *pvdL* and *tssH*, and biofilm formation genes like *algC* were particularly prevalent in sediment and droppings. Among MGEs, plasmids and transposons like Tn6072 and Tn4001 were the most abundant, playing a critical role in horizontal gene transfer. Bacterial genera including *Bacteroides*, *Clostridium*, and *Escherichia* were strongly associated with MGEs, indicating their role in the dissemination of resistance and virulence traits. Statistical analyses revealed significant differences in the abundance of ARGs, VFGs, and MGEs across sample types, with sediment and droppings identified as hotspots for gene exchange. These findings underscore the urgent need for improved antibiotic stewardship and waste management practices to limit the spread of antimicrobial resistance and pathogenic bacteria within integrated farming environments.

Keywords Antimicrobial resistance genes, Virulence factors, Mobile genetic elements, Integrated farming systems, Horizontal gene transfer

The intensification of agricultural practices, particularly within integrated farming systems, has emerged as a pivotal strategy to meet the escalating global demand for food¹. Integrated chicken-fish farming systems, prevalent in Bangladesh, exemplify an approach that enhances resource optimization and agricultural productivity². However, such systems also pose critical public health challenges, primarily linked to the propagation of antimicrobial resistance genes (ARGs)³. The indiscriminate and excessive use of antibiotics in livestock sectors, notably in poultry and aquaculture, has been identified as a principal catalyst for the emergence and proliferation of antimicrobial resistance (AMR)⁴. This phenomenon has facilitated the selection and dissemination of resistant bacterial strains, exacerbating the global AMR crisis—a threat that the World Health Organization (WHO) has ranked among the top ten public health concerns worldwide⁵.

In integrated farming systems, the close interaction between terrestrial and aquatic environments creates unique conditions that can facilitate the transfer and dissemination of ARGs⁶. Shared resources, such as water and feed, coupled with the recycling of animal waste, create conditions conducive to the spread of ARGs across microbial communities⁷. Mobile genetic elements (MGEs), such as plasmids, transposons, and insertion

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sequences (IS elements), play a critical role in this process by facilitating the horizontal gene transfer (HGT) of ARGs, further compounding the risks posed by antimicrobial resistance^{8,9}. Virulence factors, which contribute to bacterial pathogenicity, are also a concern in these interconnected systems, as they can exacerbate the impact of resistant infections¹⁰. The presence of virulence factors alongside antimicrobial resistance genes can increase the pathogenic potential of bacterial strains, heightening the risk of severe infections that are more difficult to treat¹¹.

Despite increasing awareness of the risks linked to integrated farming systems, a substantial knowledge gap persists regarding the prevalence and dissemination of antimicrobial resistance genes (ARGs), mobile genetic elements (MGEs), and virulence factors within these environments. This issue is particularly alarming in regions such as South Asia, where integrated farming practices are pervasive, and agricultural antibiotic use remains largely unregulated¹². The potential for these systems to act as hotspots for the emergence and spread of AMR requires urgent attention.

Recent advancements in metagenomic technologies have provided powerful tools for studying complex microbial communities and their associated resistomes and virulomes¹³. Metagenomic approaches enable the detection of both culturable and non-culturable organisms, offering a comprehensive overview of the microbial diversity within integrated farming systems¹⁴. By identifying the presence of ARGs and virulence genes, as well as their potential for dissemination through MGEs, these analyses provide critical insights into the public health risks posed by these agricultural practices.

Given the significant implications of ARGs and virulence factors for global health, it is essential to investigate their dissemination within integrated farming systems. Understanding the pathways through which ARGs and virulence genes spread can help develop strategies to mitigate these risks¹⁵. This study explores the ARGs, Virulence Genes and Mobile Genetic Elements (MGEs) within integrated chicken and fish farming systems in Bangladesh, focusing on the potential for these systems to serve as reservoirs and transmission hubs. The findings of this research will contribute to broader efforts to combat AMR and inform policies aimed at promoting sustainable and safe agricultural practices¹⁶.

Result

Abundance of ARGs across samples

Metagenomic analysis identified a diverse array of Antimicrobial Resistance Genes (ARGs) across the entire integrated chicken and fish farming system. In total, 384 distinct resistance genes were detected, indicating the potential for antimicrobial resistance to emerge throughout the farming system.

All the identified genes belonged to 40 different resistance classes where Tetracyclines were the most prevalent AMR (Antimicrobial Resistance) class, comprising 20.4% of the total resistance genes detected. This was followed by macrolide-lincosamide-streptogramin (17.6%) and aminoglycosides (15%). Droppings accounted for the highest proportion of AMR genes, contributing 62.2% of the total, while Sediment accounted for 31.5%. Chicken Gut, Feed, and Fish Intestine accounted for 5.3%, 0.9%, and 0.1%, respectively. Other AMR classes were detected at lower frequencies, including multi-drug resistance (4.3%), sulfonamides (3.9%), rifampin (3.8%), and phenicol (2.9%). Classes related to biocide and metal resistance, such as drug and biocide resistance (9.5%) and multi-metal resistance (2.5%), were also observed. Additionally, resistance to beta-lactams (1.4%), glycopeptides (0.4%), and fluoroquinolones (2.9%) was detected, albeit at lower levels (Fig. 1).

In terms of analyzing sample-wise relative abundance of ARGs, it was found that in the Feed sample, the most prevalent AMR classes were macrolide-lincosamide-streptogramin (28.60%), aminoglycosides (20.19%), and oxazolidinone (17.97%). Notable, though less abundant, were cationic antimicrobial peptides (6.00%) and fluoroquinolones (6.15%). The Chicken Gut sample demonstrated a dominant presence of macrolide-lincosamide-streptogramin (33.23%) and tetracyclines (22.07%), followed closely by aminoglycosides (19.12%). Sulfonamides (5.96%) and phenicol (5.29%) were also present, suggesting a complex resistance profile in the gut microbiome. Droppings exhibited the highest concentration of tetracyclines (28.43%) among all samples, with significant levels of macrolide-lincosamide-streptogramin (19.41%) and multi-drug resistance (9.63%). The presence of phenicol (5.96%) and sulfonamides (5.96%) indicates a broad spectrum of resistance within this sample. In the Fish Intestine sample, oxazolidinone (13.55%) and macrolide-lincosamide-streptogramin (14.53%) were the predominant AMR classes, with aminoglycosides (10.84%) also contributing notably. Fluoroquinolones (7.88%) were observed at lower levels, indicating a distinct resistance profile compared to other samples. The Sediment sample displayed a unique pattern, with multi-metal resistance (34.73%) being the most prevalent, followed by macrolide-lincosamide-streptogramin (10.95%) and aminoglycosides (10.99%). The significant presence of rifampin (9.11%) and phenicol (5.95%) further highlighted the diversity of resistance genes in this environmental reservoir (Fig. 2A).

Since resistance was detected across all samples within the farming system, statistical analyses were conducted to assess whether significant differences existed among the samples in terms of ARG presence. To address this, the distribution of antimicrobial resistance genes (ARGs) was analyzed using the F-Welch test, which is robust against unequal variances and sample sizes. The F-Welch test revealed a significant overall difference in ARG abundance among the samples [$F_{Welch}(4, 51.76) = 7.60, p = 6.77e-05$]. To further explore these differences, the Games-Howell pairwise analysis was conducted, which also adjusts for unequal variances. Games-Howell test showed statistically significant differences with the effect size ($\omega^2_p = 0.42, CI_{95\%} [0.31, 1.00]$), indicating that the magnitude of these differences are high. The Bayesian analysis provided a $\log_e(BF_{01})$ of -0.41, with a posterior probability of 0.05 and a 95% credible interval [0.00, 0.15], along with an rCauchy (Random Cauchy) JZS (Jeffreys-Zellner-Siow) value of 0.71 further support the result (Fig. 2B).

To identify the most abundant ARGs, 30 most abundant genes are represented in Table 1 along with their class, function and abundance.

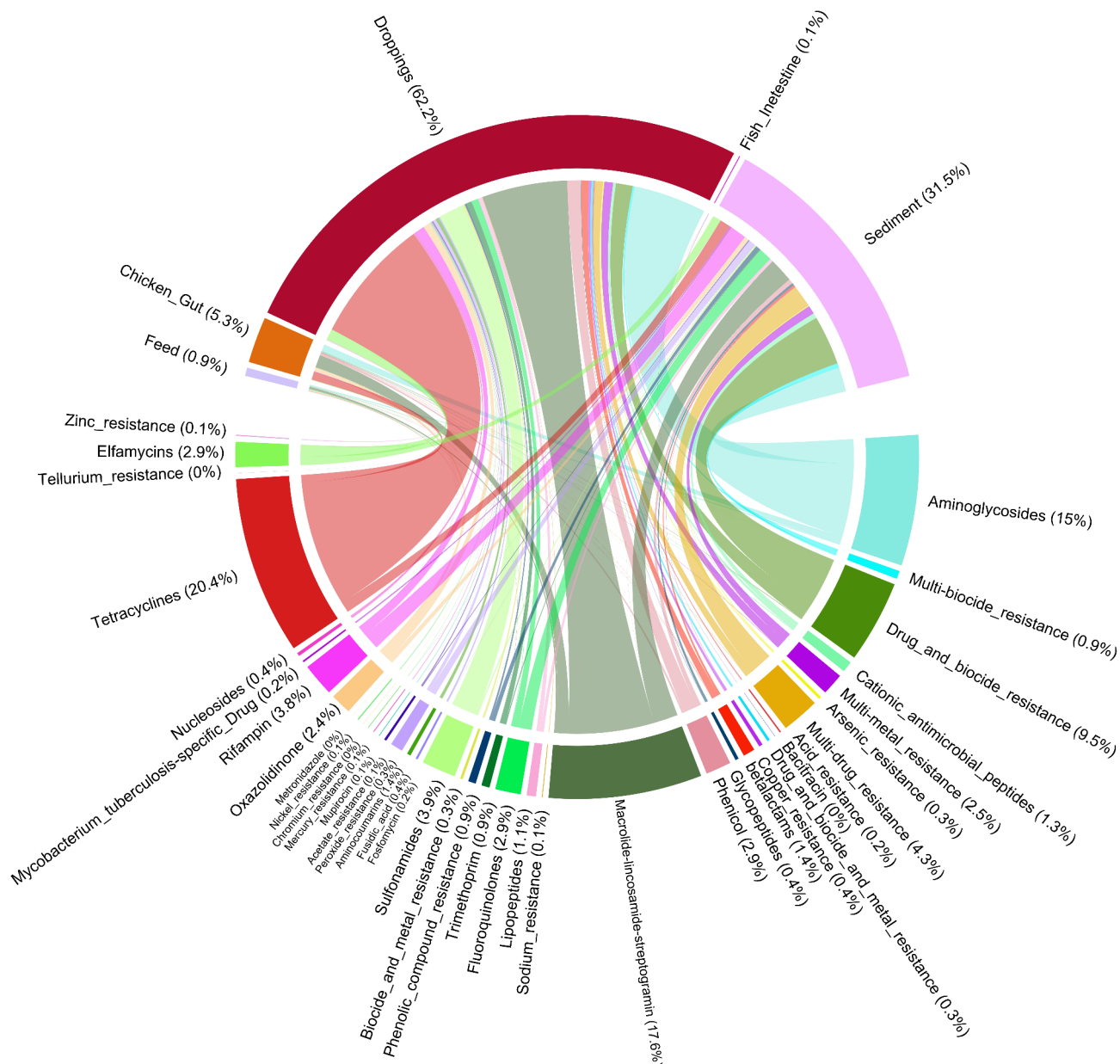
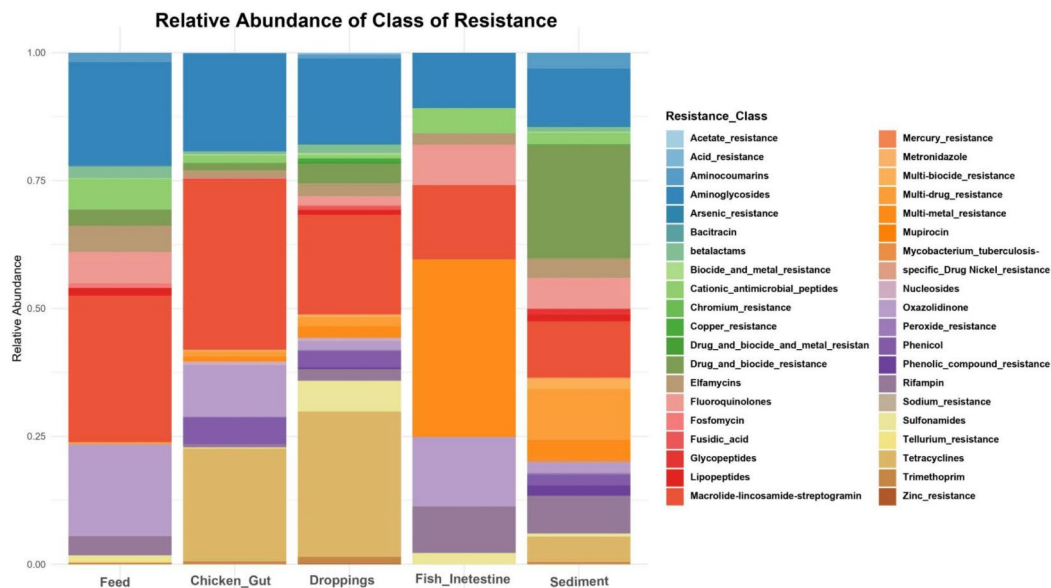


Fig. 1. Chord diagram illustrating the distribution of antimicrobial resistance genes (ARGs) across different sample types. Droppings harbored the highest proportion of ARGs, followed by Sediment. The most prevalent resistance class was Tetracyclines, with macrolide-lincosamide-streptogramin and aminoglycosides also showing significant presence.

Virulence factors

A total of 445 types of virulence factor-associated genes were identified, belonging to 12 different mechanism classes. Among these, immune modulation was most abundant in Fish Intestine (67.01%) and Sediment (3.07%). The effector delivery system was highest in Chicken Gut (33.77%) and Sediment (34.82%). biofilm formation showed the greatest abundance in Feed (27.52%), followed by Chicken Gut (11.4%) and Droppings (9.55%). Motility was most prevalent in Droppings (20.1%) and Chicken Gut (13.4%). Other mechanisms such as Adherence were prominent in Droppings (18.05%) and Chicken Gut (14.43%). Nutritional/Metabolic factors were more abundant in Sediment (23.63%) and Droppings (14.03%). Exotoxin was mainly present in Fish Intestine (5.56%), and invasion was detected only in Droppings (1.54%). Mechanisms such as exoenzyme and Nutritional/Metabolic factors showed minimal presence across samples (Fig. 3A).

There were significant differences in virulence factor abundance among the different sample types. Sediment had significantly higher virulence factor abundance compared to Feed (pHolm-adj. = 6.43e-13), Chicken Gut (pHolm-adj. = 5.88e-12), Fish Intestine (pHolm-adj. = 7.20e-12), and Droppings (pHolm-adj. = 1.71e-12). Droppings also showed significant differences compared to Feed (pHolm-adj. = 0.01) and Chicken Gut (pHolm-adj. = 0.00). Additionally, Chicken Gut exhibited significant differences when compared to Fish Intestine

A**B**

Distribution of ARGs Across Samples

$F_{Welch}(4, 136.97) = 26.57, p = 2.57e-16, \hat{w}_p^2 = 0.42, CI_{95\%} [0.31, 1.00], n_{obs} = 384$

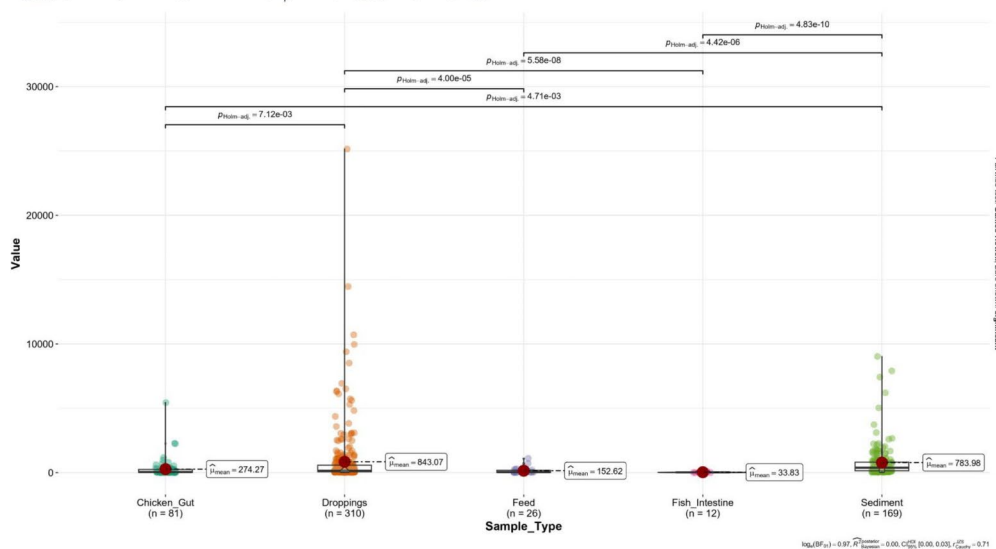


Fig. 2. (A) Relative abundance plot showing the sample-wise distribution of antimicrobial resistance genes (ARGs). In the Feed sample, the most prevalent AMR classes included macrolide-lincosamide-streptogramin, aminoglycosides, and oxazolidinone. The Chicken Gut sample demonstrated dominance of macrolide-lincosamide-streptogramin and tetracyclines, while Droppings exhibited the highest concentration of tetracyclines. Sediment displayed a unique profile with multi-metal resistance being the most prevalent class. (B) Bar plot illustrating the statistical analysis of antimicrobial resistance gene (ARG) abundance across different samples. The F-Welch test indicated a significant overall difference in ARG abundance among the samples. Subsequent Games-Howell pairwise analysis revealed statistically significant differences with a high effect size, suggesting substantial variability in resistance levels. Bayesian analysis further supported these findings, reinforcing the robustness of the observed differences.

Genes	Description	Mechanism	Droppings	Sediment
<i>pvdL</i>	Peptide synthase PvdL Pyoverdine	Nutritional/Metabolic factor	0%	100%
<i>pvdI</i>	Peptide synthase Pyoverdine	Nutritional/Metabolic factor	0%	100%
<i>chpA</i>	Signal transduction system Type IV pili	Adherence	0%	100%
<i>pchF</i>	Pyochelin synthetase PchF Pyochelin	Nutritional/Metabolic factor	0%	100%
<i>clpV1</i>	Type VI secretion system AAA + family ATPase HSI-1	Effector delivery system	32.38%	67.62%
<i>ppkA</i>	Serine/threonine protein kinase PpkA HSI-1	Effector delivery system	28.11%	71.89%
<i>tssH-5/clpV</i>	Clp-type ATPase chaperone protein T6SS-1	Effector delivery system	0%	100%
<i>tssH</i>	Type VI secretion system ATPase TssH HSI-2	Effector delivery system	35.72%	64.28%
<i>pchE</i>	Dihydroaeruginic acid synthetase PchE Pyochelin	Nutritional/Metabolic factor	0%	100%
<i>algC</i>	Phosphomannomutase AlgC Alginate	Biofilm	34.03%	65.97%
<i>pvdD</i>	Pyoverdine synthetase D Pyoverdine	Nutritional/Metabolic factor	0%	100%
<i>icmF1/tssM1</i>	Type VI secretion system protein IcmF1 HSI-1	Effector delivery system	0%	100%
<i>phzE1</i>	Phenazine biosynthesis protein PhzE Pyocyanin	Nutritional/Metabolic factor	27.99%	72.01%
<i>pilJ</i>	Twitching motility protein PilJ Type IV pili	Adherence	37.59%	62.41%
<i>flhA</i>	Flagellar biosynthesis protein FlhA Flagella	Motility	38.95%	61.05%
<i>pvdJ</i>	Pyoverdine biosynthesis protein PvdJ Pyoverdine	Nutritional/Metabolic factor	0%	100%
<i>tse5/rhsP1</i>	Toxin protein Tse5 HSI-1 T6SS secreted effectors	Effector delivery system	0%	100%
<i>xcpR</i>	General secretion pathway protein E Xcp secretion system	Effector delivery system	32.44%	67.56%
PA1458	Two-component sensor Flagella	Motility	33.4%	66.6%
<i>cheA</i>	Chemotaxis two-component sensor kinase CheA Flagella	Motility	0%	100%
<i>tsr</i>	Methyl-accepting chemotaxis protein I Flagella	Motility	0%	100%
<i>pilT</i>	Twitching motility protein PilT Type IV pili	Adherence	41.06%	58.94%
<i>fleQ</i>	Transcriptional regulator FleQ Flagella	Motility	32.82%	67.18%
<i>tssM</i>	Type VI secretion system membrane subunit TssM HSI-2	Effector delivery system	0%	100%
<i>fimX</i>	Protein FimX Type IV pili	Adherence	30.31%	69.69%

Table 1. 30 Most abundant antimicrobial resistance genes (ARGs) with their class, functions and abundance. *Top 25 Virulence factors only appeared in Droppings and Sediment.

($p_{\text{Holm-adj.}} = 0.03$). The number of virulence factors present in each sample were: Chicken Gut ($n = 39$), Droppings ($n = 206$), Feed ($n = 32$), Fish Intestine ($n = 26$), and Sediment ($n = 314$) (Fig. 3B).

The top 25 most abundant virulence genes were detected exclusively in Droppings and Sediment samples, with no presence in Feed, Chicken Gut, or Fish Intestine samples. Certain genes, including *pvdL*, *pvdI*, *pchF*, *tssH-5/clpV*, *pvdJ*, *icmF1/tssM1*, *cheA*, *tssM*, and *tsr*, were found solely in Sediment with 100% abundance. Other genes displayed varying distributions between Droppings and Sediment. For instance, *clpV1* (32.38% in Droppings, 67.62% in Sediment), *ppkA* (28.11% in Droppings, 71.89% in Sediment), *pilT* (41.06% in Droppings, 58.94% in Sediment), *phzE1* (27.99% in Droppings, 72.01% in Sediment), *algC* (34.03% in Droppings, 65.97% in Sediment), and *pilJ* (37.59% in Droppings, 62.41% in Sediment) showed higher abundance in Sediment than in Droppings. Motility-related genes like *flhA* (38.95% in Droppings, 61.05% in Sediment) and *fleQ* (32.82% in Droppings, 67.18% in Sediment) followed a similar pattern, favoring Sediment (Table 2).

Abundance of mobile genetic elements

In the integrated chicken and fish farming system, a total of 2887 mobile genetic elements (MGEs) were identified, divided into six distinct types: Plasmids ($n = 1,912$), Insertion Sequences (IS) ($n = 882$), Integrative and Conjugative Elements (ICE) ($n = 39$), Transposons (Tn) ($n = 26$), Chromosomal Islands (CN) ($n = 18$), and Integrative Mobilizable Elements (IME) ($n = 10$). Plasmids, which are autonomous extrachromosomal DNA molecules, were the most prevalent, while IMEs, responsible for genome integration and mobilization, were the least common.

A Kruskal-Wallis test revealed significant overall differences between the six MGE types ($\chi^2(5) = 115.01$, $p = 3.58 \times 10^{-23}$). Pairwise comparisons using Dunn's test with Holm adjustment showed several significant differences. Notably, ICE and IME differed significantly from other types, with p -values as low as 1.21×10^{-14} . Additionally, IS and Plasmids, Tn and ICE, CN and IME, and other pairings showed significant variations with adjusted p -values ranging from 8.07×10^{-4} to 6.30×10^{-17} . These results highlight a complex and distinct distribution of MGE types, reflecting their varied roles in horizontal gene transfer and ecological adaptation within the farming system (Fig. 4A).

Abundance of (MGEs) were identified across five sample types in the farming system counting as: Droppings ($n = 1,823$), Sediment ($n = 1,321$), Chicken Gut ($n = 465$), Feed ($n = 227$), and Fish Intestine ($n = 141$). Droppings and Sediment exhibited the highest abundance of MGEs, while the Fish Intestine had the lowest.

A Kruskal-Wallis test indicated significant overall differences in MGE abundance among the sample types ($\chi^2(4) = 261.61$, $p = 2.06 \times 10^{-55}$). Pairwise comparisons using Dunn's test with Holm adjustment revealed several

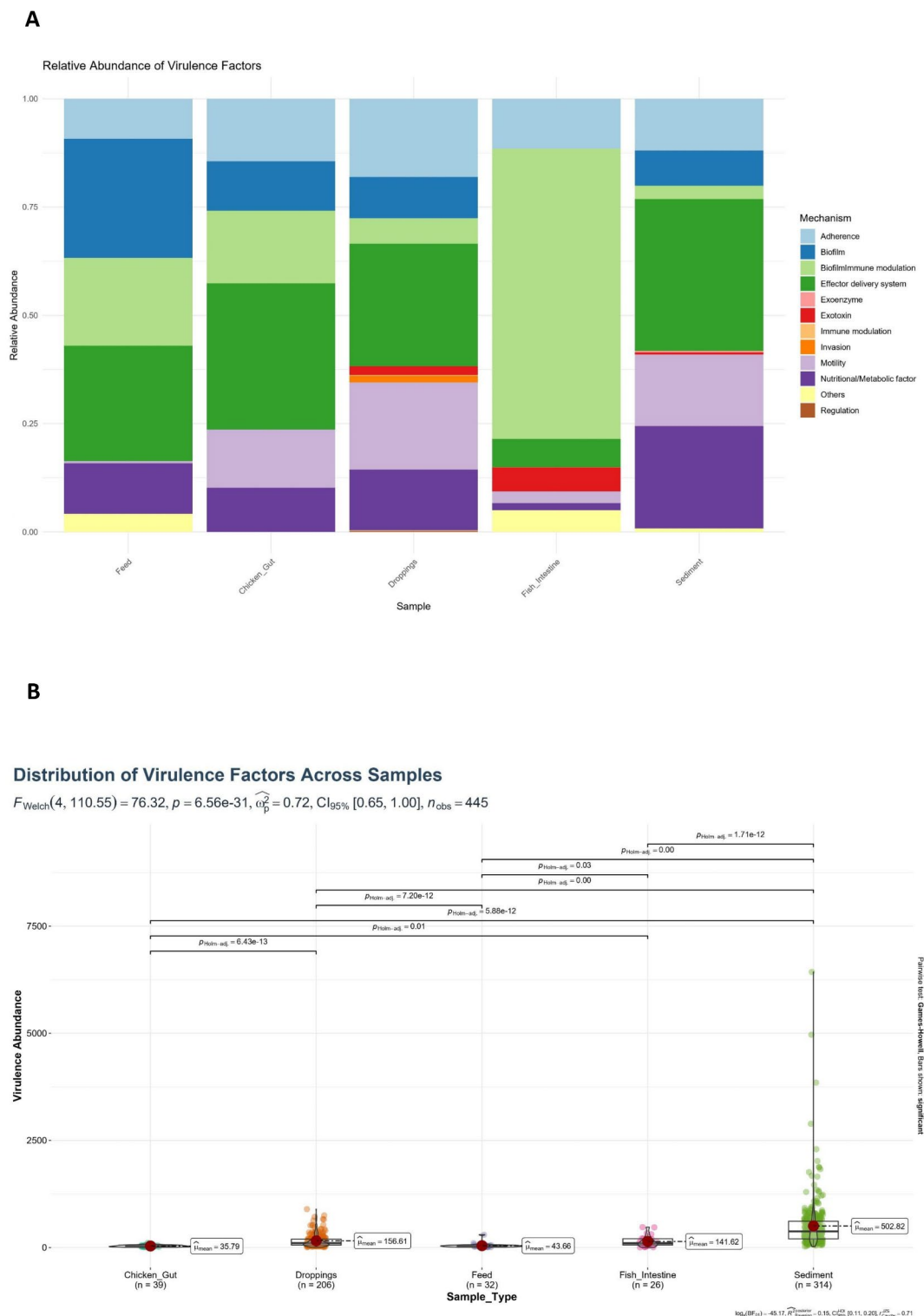


Fig. 3. (A) Relative abundance plot depicting the distribution of virulence factor-associated genes across different samples. A total of 445 types of virulence factor-associated genes were identified, with immune modulation being most abundant in Fish Intestine and Sediment. The effector delivery system was highest in Chicken Gut and Sediment, while biofilm formation was predominant in Feed. Additionally, motility and adherence mechanisms were notable in Droppings and Chicken Gut, highlighting the diverse virulence profiles across samples. (B) Bar plot illustrating the abundance of virulence factor-associated genes across different sample types. Significant differences in virulence factor abundance were observed among the samples, with Sediment exhibiting the highest count compared to Feed, Chicken Gut, Fish Intestine, and Droppings. Droppings also showed significant differences when compared to Feed and Chicken Gut, while Chicken Gut had notable differences compared to Fish Intestine. The number of virulence factors identified in each sample varied, highlighting the diverse virulence profiles present.

MGE	Type	Genera	ARGs	Virulence Genes
ICESGB76	ICE	12	LSAE, CAT, RPSL, LSA, TUFAB, FLOR, RRSA, LNUB	<i>xcpR</i> , <i>xcpA/pilD</i> , <i>wzy</i> , <i>tssC</i> , <i>tssB</i> , <i>tse7</i> , <i>tse1</i> , <i>rpoN</i> , <i>pv dF</i>
GISul2	IME	13	CATB, ERM, FOLP, BRP, ERED, APH4, ERMG, TETA, QACL	<i>waaP</i> , <i>waaG</i> , <i>vfr</i> , PA1464, PA1459, PA1458, <i>mucE</i> , <i>mucA</i>
Tn558	Tn	11	DFRG, TET16S	<i>xcpU</i> , <i>xcpR</i> , <i>xcpA/pilD</i> , <i>wzy</i>
Tn4351	Tn	13	ERM, FOLP, BRP, ERED, APH4, ERMG, TETA, QACL, APH6, ERMF, OXA	<i>tssC</i> , <i>tssB</i> , <i>tse7</i> , <i>tse1</i> , <i>rpoN</i> , PA1458, <i>mucE</i> , <i>mucA</i>
Tn6031	Tn	13	ERM, FOLP, BRP, ERED, APH4, ERMG, TETA, QACL, APH6, ERMF, OXA, DFRA, MEFA	<i>xcpU</i> , <i>xcpR</i> , <i>xcpA/pilD</i> , <i>wzy</i>
Tn6191	Cn	12	0	PA1458, <i>mucE</i> , <i>mucA</i>
Tn6246	Cn	12	DFRE	<i>tse7</i> , <i>tse1</i> , <i>rpoN</i> , PA1458
Tn5432	Cn	13	APH4, APH6, BLE, ERMT, DFRE, APH3-PRIME, ANT6, APH3PRIME, ERMX, MYRA, LNUC, TETZ	<i>waaP</i> , <i>waaG</i> , <i>vfr</i> , PA1464
Tn4001	Cn	11	BLE, ERMT, DFRE, MYRA, LNUC, TETZ, CFRA, CMX	<i>tssC</i> , <i>tssB</i>
Tn2017	Tn	11	APH3-PRIME, ANT6, APH3-PRIME, ERMX, LNUC, TETX, AAC3, DHFR, QACEDELTA1, ANT9, SULI, SULI, E RMB, ANT3-PRIME, AAC6-PRIME, TETQ, CFRA	0
CTnBST	ICE	11	APH4, APH6, BLE, ERMT	0
ICEBsa18170-1	ICE	10	DFRG, TET16S	0
ICEBfrYCH46-1	ICE	10	DFRE	PA1458, <i>mucE</i> , <i>mucA</i>
tISCpe8	IS	5	LSAE, CAT, RPSL	<i>xcpR</i> , <i>tssC</i>
CTn341	ICE	8	BLE, ERMT, DFRE, MYRA, LNUC, TETZ, CFRA	<i>vfr</i> , PA1464
Tn6105	IME	8	CATB, ERM, FOLP, BRP, ERED, APH4	<i>tssC</i>
Tn552	Tn	7	DFRG, TET16S	PA1458
Tn6106	IME	7	LSAE, CAT, RPSL	<i>tssC</i>
Tn6061	Tn	4	DFRE	0

Table 2. Distribution of the top 25 most abundant virulence genes detected in droppings and sediment samples.

significant differences, particularly between Droppings and other sample types, with p -values as low as 3.80e-38. Additional significant differences were observed between Sediment and Feed ($p = 7.84e-12$), Chicken Gut and Fish Intestine ($p = 1.91e-16$), and multiple other comparisons, with adjusted p -values ranging from 1.12e-03 to 4.06e-17. These findings demonstrate distinct patterns of MGE distribution across the different environmental samples, suggesting varied levels of gene exchange and microbial activity (Fig. 4B).

Correlation among organisms, ARGs and virulence genes

The correlation analysis between the top 25 most abundant MGEs and ARGs, using Spearman correlation with a threshold of $r > 0.7$ and $p < 0.05$, reveals several strong positive correlations. MGEs such as Tn6072, Tn6246, and Tn4001 exhibit significant positive correlations with ARGs like *tetM*, *sulI*, and *tetX*. Similarly, ICEBfrYCH46-1 shows a strong positive correlation with *tetW* and *gyrA*, while Tn6191 correlates positively with *tetQ*. Other notable positive correlations include CTn341 with *rpoB*, ISLjo5 with *tetL*, and CTnHyb with *sulII*. These results indicate a widespread co-association between MGEs and ARGs, suggesting the potential role of MGEs in the dissemination of antibiotic resistance within microbial communities (Fig. 5A).

The correlation analysis between the top 25 virulence genes revealed that the transposon element Tn125 and the plasmid *Nonomuraea gerenzanensis* isolate nono1 plasmid I showed a positive correlation with multiple virulence genes, including *pvdL*, *pvdI*, *chpA*, *pchF*, *tssH-5/clpV*, *pchE*, *pvdD*, *icmF1/tssM1*, *pvdI*, *tse5/rhsP1*, *cheA*, *tsr*, and *tssM* (Fig. 5B).

A Spearman correlation analysis was conducted between microbial taxa and Mobile Genetic Elements (MGEs), focusing on the top 700 most abundant Operational Taxonomic Units (OTUs) out of 5796 identified OTUs. The analysis filtered results to retain only significant correlations where the correlation coefficient (r) was greater than 0.7 and the p -value was less than 0.05. A total of 521 OTUs, representing 260 taxonomic genera, were found to be significantly correlated with 453 MGEs.

The genus *Bacteroides* was strongly correlated with multiple MGEs, including ISEc36, ISKpn6, and ISVsa2. Similarly, *Clostridium* exhibited notable correlations with IS116, ISKpn26, and ISLsa11, indicating a close association with these genetic elements. The genus *Escherichia* was linked to several MGEs, such as ISEc52, ISKpn43, and ISPa42, showing its potential role in disseminating these elements across microbial communities. *Spirochaeta* demonstrated strong correlations with ISPme5 and ISXyl3, while *Actinomyces* was associated with ISXyl2 and ISEc54, highlighting their connections to distinct MGEs. Further correlations were observed between *Lactobacillus* and IS26, *Prevotella* and ISGca8, and *Streptococcus* with both IS26 and IS903, suggesting widespread genetic exchanges involving these genera. Additionally, *Bifidobacterium* was correlated with ISKpn7, and *Pseudomonas* showed a strong correlation with ISEcp1.

Moreover, *Enterococcus* was significantly correlated with IS26, IS903, and ISEc54, while *Faecalibacterium* displayed associations with IS26 and IS200, reinforcing the importance of these MGEs in microbial gene flow. Lastly, *Roseburia* was linked to ISEcp1, and *Bifidobacterium* was correlated with ISKpn7 and IS903,

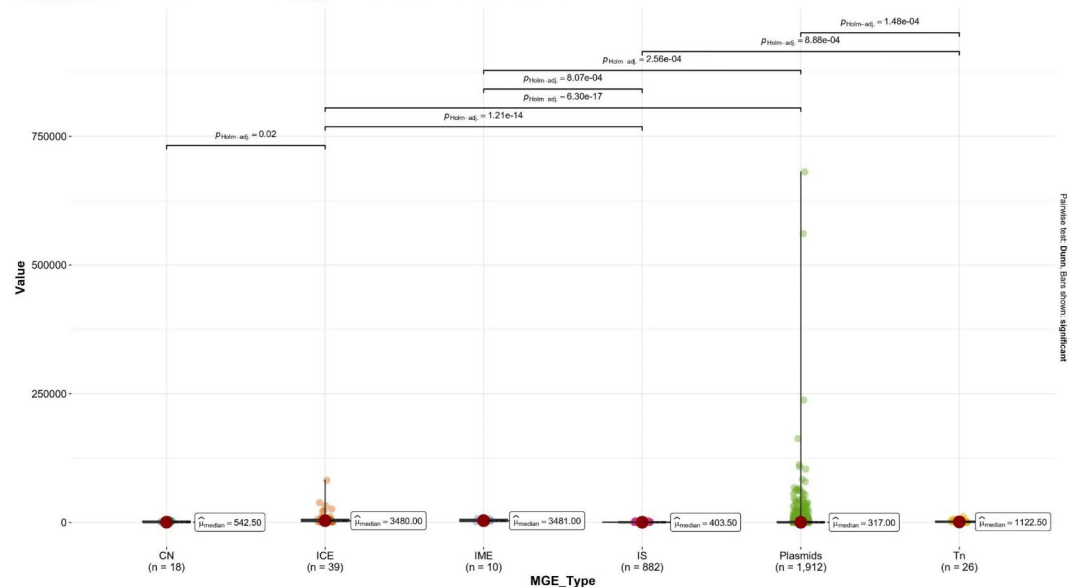
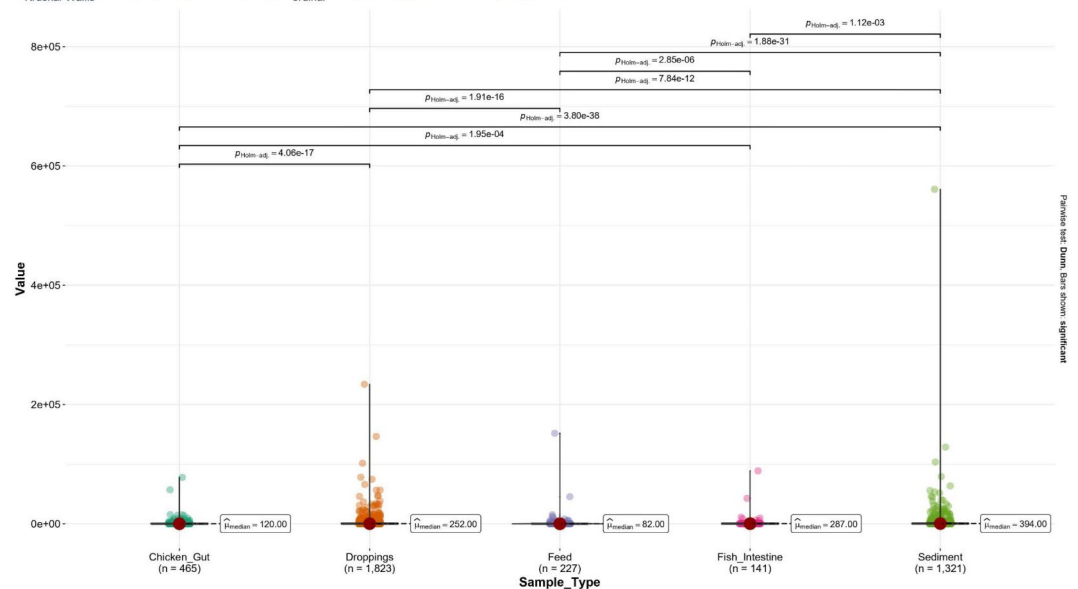
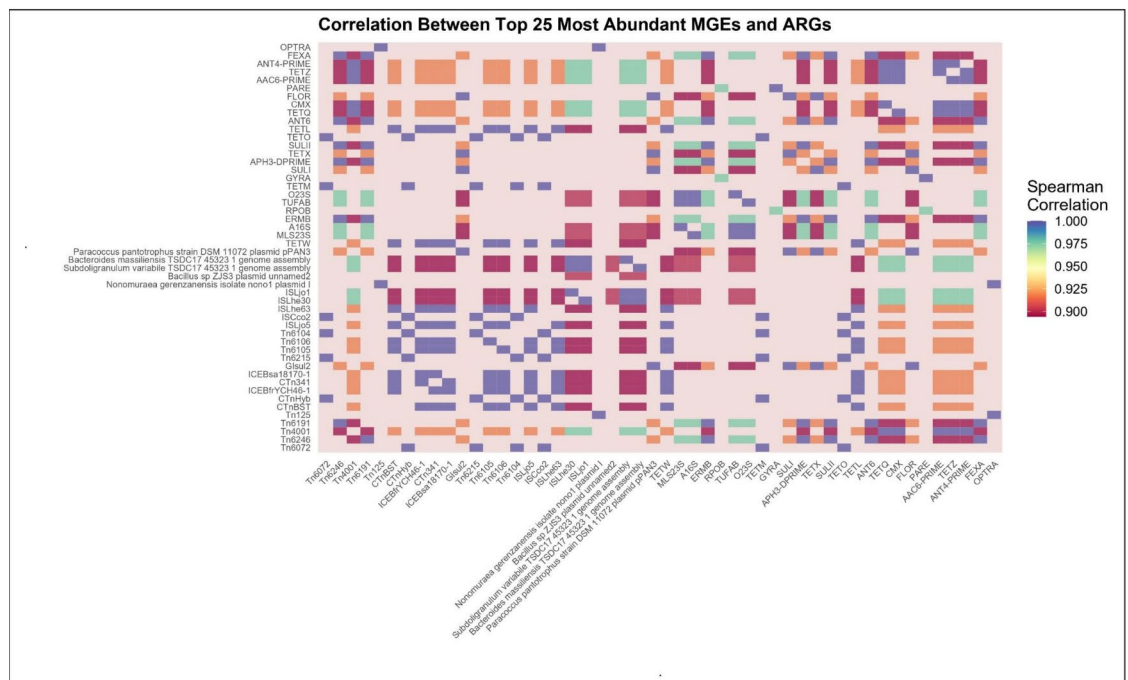
A**Distribution of MGEs**
 $\chi^2_{\text{Kruskal-Wallis}}(5) = 115.01, p = 3.58\text{e-}23, \hat{c}^2_{\text{ordinal}} = 0.04, \text{CI}_{95\%} [0.04, 1.00], n_{\text{obs}} = 2,887$
**B****Distribution of MGEs Across Samples**
 $\chi^2_{\text{Kruskal-Wallis}}(4) = 261.61, p = 2.06\text{e-}55, \hat{c}^2_{\text{ordinal}} = 0.07, \text{CI}_{95\%} [0.05, 1.00], n_{\text{obs}} = 3,977$


Fig. 4. (A) Bar plot displaying the distribution of mobile genetic element (MGE) types. A Kruskal-Wallis test indicated significant overall differences among the six MGE types. Pairwise comparisons revealed notable differences. Other significant variations were observed among IS and Plasmids, Tn and ICE, and CN and IME, emphasizing the complex roles of these elements in horizontal gene transfer and ecological adaptation within the farming system. (B) Bar plot illustrating the abundance of mobile genetic elements (MGEs) across different sample types. A Kruskal-Wallis test revealed significant overall differences in MGE abundance among the samples. Pairwise comparisons highlighted significant differences, particularly between Droppings and other sample types. Additional significant differences were observed between Sediment and Feed, as well as Chicken Gut and Fish Intestine, indicating distinct patterns of MGE distribution and suggesting varied levels of gene exchange and microbial activity in the environment.

A



B

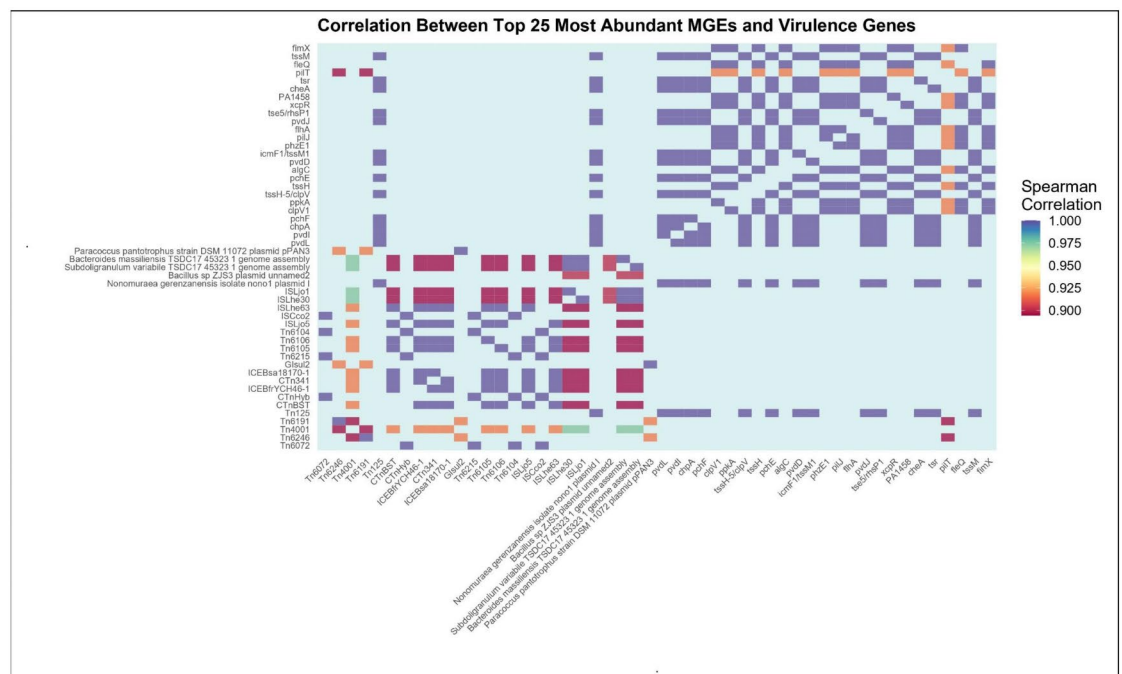
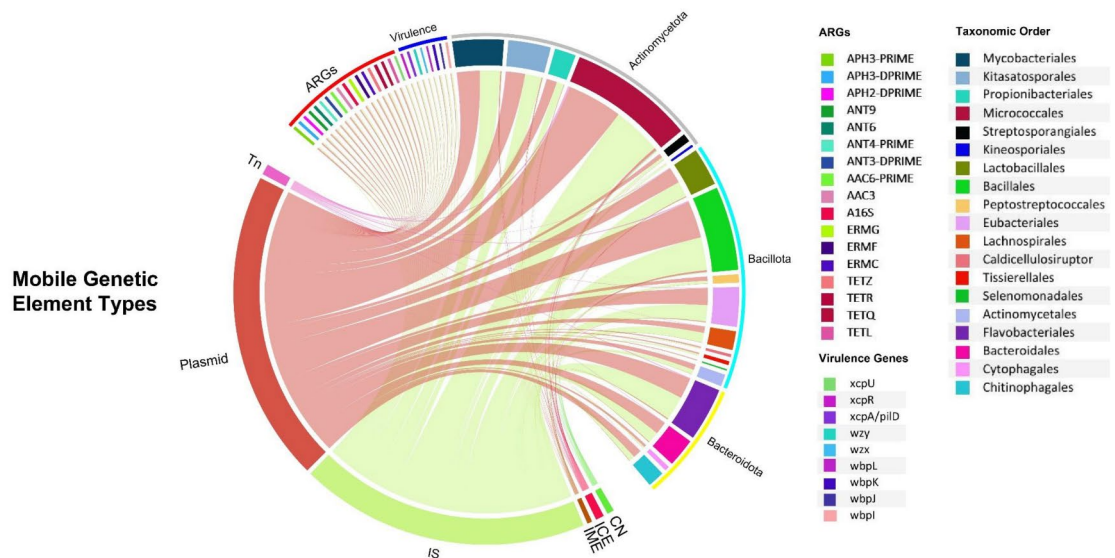
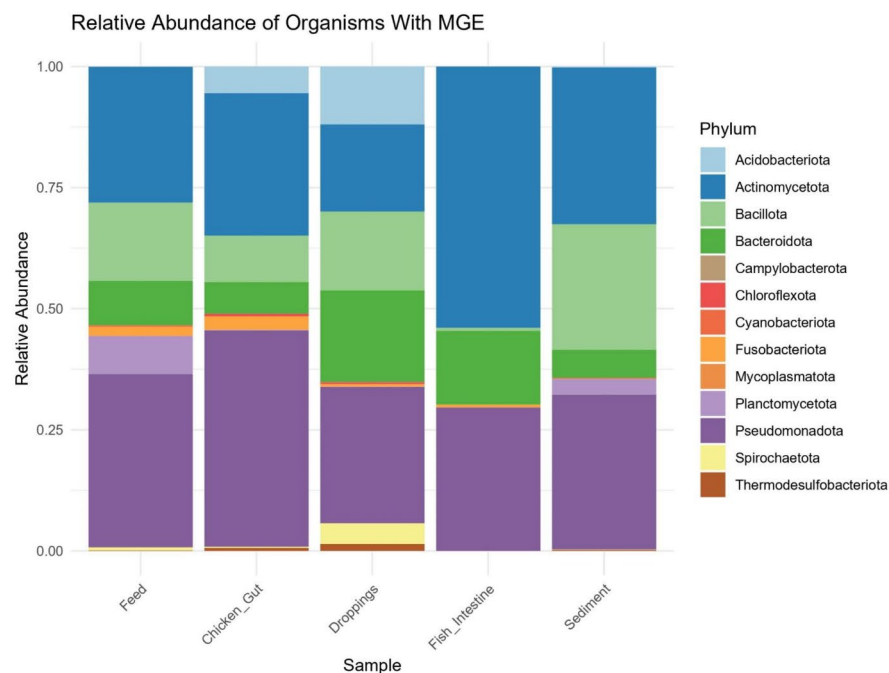


Fig. 5. (A) Heatmap depicting the correlation between the top 25 most abundant mobile genetic elements (MGEs) and antimicrobial resistance genes (ARGs). Correlation analysis using Spearman correlation revealed several strong positive correlations, with MGEs such as Tn6072, Tn6246, and Tn4001 significantly correlating with ARGs like tetM, sulI, and tetX. Additionally, ICEBfrYCH46-1 showed a strong positive correlation with tetW and gyrA, highlighting the widespread co-association between MGEs and ARGs, which suggests their potential role in the dissemination of antibiotic resistance within microbial communities. **(B)** Heatmap illustrating the correlation between the top 25 virulence genes and mobile genetic elements. Correlation analysis revealed that the transposon element Tn125 and the plasmid Nonomuraea gerezanensis isolate nono1 plasmid I exhibited positive correlations with multiple virulence genes, including *pvdL*, *pvdI*, *chpA*, and *pchF*, among others.

A



B



further demonstrating the widespread interaction between these bacterial taxa and MGEs. These correlations indicate substantial interactions between the microbial communities and MGEs, contributing to the potential dissemination of genetic material. Figure 6A, alongside of highlighting correlation between MGEs and organisms, also highlights the correlation among most significant ($p < 0.01$) ARGs and Virulence genes with the MGEs.

The relative abundance of microbial phyla associated with MGEs shows distinct patterns in terms of presence throughout the whole farming system. Pseudomonadota was the most abundant, especially in Chicken_Gut (44.62%) and Fish_Intestine (29.62%). Actinomycetota was prevalent in Feed (28.05%) and Sediment (32.37%), while Bacillota was abundant in Sediment (25.93%) and Droppings (16.26%). Bacteroidota was notable in Droppings (18.92%) and Fish_Intestine (15.25%). Thermodesulfobacteriota, Spirochaetota, and other phyla showed lower abundance, with Mycoplasmatota and Campylobacterota nearly absent across all samples (Fig. 6B).

◀ **Fig. 6.** (A) Chord diagram illustrating the correlations between mobile genetic elements (MGEs) and various microbial genera, along with significant antimicrobial resistance genes (ARGs) and virulence genes. The analysis reveals strong correlations between the genus *Bacteroides* and multiple MGEs, such as ISEc36 and ISKpn6, while *Clostridium* showed notable associations with IS116 and ISKpn26. *Escherichia*, *Spirochaeta*, and *Actinomyces* also exhibited significant links to various MGEs, indicating their roles in genetic dissemination. The diagram further highlights the interactions between these bacterial taxa and significant ARGs and virulence genes, underscoring the potential for widespread genetic exchange within microbial communities. (B) Relative abundance plot illustrating the distribution of microbial phyla associated with mobile genetic elements (MGEs) across different samples. Pseudomonadota was the most abundant phylum, particularly in Chicken Gut and Fish Intestine. Actinomycetota exhibited high prevalence in Feed and Sediment, while Bacillota was notably abundant in Sediment and Droppings. Bacteroidota was also significant in Droppings and Fish Intestine. Other phyla, such as Thermodesulfobacteriota and Spirochaetota, showed lower abundance, with Mycoplasmatota and Campylobacterota being nearly absent across all samples.

The correlation analysis revealed a significant association between various Mobile Genetic Elements (MGEs), bacterial genera, and their capacity to disseminate antimicrobial resistance genes (ARGs) and virulence genes. Among the most abundant MGEs, ICE SGB76 was linked to 12 genera and associated with ARGs such as *lsa*, *eca*, and *tuf*, alongside virulence genes like *xcpR* and *tssC*. Similarly, IME GISul2 was correlated with 13 genera, disseminating ARGs like *cat*, *ber*, and *gtet*, as well as virulence genes such as *waaP* and *mucA*. Transposon Tn558 was found to be connected to 11 genera, spreading ARGs like *dfr* and *gtet* and virulence genes such as *xcpU* and *pilD*. Tn4351 and Tn6031, both linked to 13 genera, were identified to disseminate various ARGs including *erm* and *fol*, with virulence genes like *tssC* and *xcpA*. Conjugative elements like Tn6191 and Tn6246 were associated with 12 genera, transmitting virulence genes such as *mucA* and *rpoN*. The study also highlighted Tn5432 and Tn4001, which are linked to the spread of ARGs such as *aph* and *ble* across 13 and 11 genera, respectively. Other significant MGEs, such as ICEBs18170-1, tISCpe8, and CTn341, were also associated with the transfer of ARGs and virulence genes, indicating their crucial role in microbial gene dissemination (Table 3).

Discussion

The metagenomic analysis of antimicrobial resistance genes (ARGs) across the integrated chicken and fish farming system highlights the complex interplay between agricultural practices and the dissemination of resistance genes. The widespread detection of ARGs across different samples reflects the pervasive nature of resistance in such environments, with implications for both environmental and public health¹⁷.

The dominance of tetracycline resistance genes, particularly in droppings, is indicative of the selective pressures from tetracycline use in animal farming¹⁸. Tetracyclines have long been used as growth promoters and for disease prevention in poultry, which has led to their persistence in the environment and subsequent spread through manure and other waste products¹⁹. The detection of these genes in sediment further emphasizes the role of environmental reservoirs in maintaining and disseminating ARGs, as sediments often serve as sinks for pollutants, including antibiotics and resistance genes, especially in integrated farming systems where water flow connects different environments^{20,21}.

The presence of macrolide-lincosamide-streptogramin and aminoglycoside resistance genes across various samples, particularly in the chicken gut and feed, suggests ongoing exposure to these antibiotic classes²². Macrolides and aminoglycosides are commonly used in livestock farming, and their resistance genes can be readily transferred between bacteria through horizontal gene transfer²³. The detection of oxazolidinone resistance genes, though less common, raises concerns as oxazolidinones are often considered last-resort antibiotics in human medicine, and the presence of resistance genes in the farming environment could contribute to the diminishing efficacy of these drugs^{24,25}.

Among the detected ARGs, those encoding resistance to last-resort antibiotics such as, *TETW*, *MLS23S*, *RPOB*, and *GYRA* belong to high-risk antimicrobial resistance genes (ARGs)^{26–28}. These genes are linked to resistance against tetracyclines, macrolides, rifampin, and fluoroquinolones, respectively, and pose significant threats to both public and veterinary health^{6,18}. When expressed at high levels in animal environments, such as feed, droppings, and sediment, these genes can act as reservoirs for resistance, potentially transferring to humans either directly through consumption or indirectly via environmental exposure²⁹. Medium-risk ARGs, such as *ERMB* and *PARE*, also warrant attention despite their less widespread expression^{30,31}. *ERMB*, involved in macrolide resistance, has been detected in animal environments, indicating a potential for resistance spread³². Similarly, *PARE*, related to aminocoumarin resistance, could contribute to DNA topoisomerase resistance, though its impact is still emerging³³. These genes, though not as pervasive as their high-risk counterparts, still contribute to the broader AMR issue.

The identification of biocide and multi-metal resistance genes, particularly in sediment, highlights the role of environmental contaminants beyond antibiotics in selecting for resistance³⁴. Heavy metals and biocides are often co-selectors for antibiotic resistance³⁵, as bacteria that develop resistance to one type of stressor may simultaneously acquire resistance to others³⁶. This co-selection process complicates efforts to control the spread of ARGs, as it suggests that reducing antibiotic use alone may not be sufficient to curb resistance in such complex environments³⁷.

While this study provides insights into the presence of ARGs across various environmental samples, it is important to consider that the mere detection of ARGs does not confirm their expression or activity. Recent findings³⁸ indicate that a substantial proportion of ARGs in animal and human gut microbiota are transcriptionally inactive, which underscores the need for integrative metagenomic and metatranscriptomic

Gene	Class	Function	Feed (%)	Chicken Gut (%)	Droppings (%)	Fish Inetestine (%)	Sediment (%)
TETW	Tetracyclines	Tetracycline_resistance_ribosomal_protection_proteins	0	8.34	91.66	0	0
MLS23S	Macrolide-lincosamidestreptogramin	Macrolide-resistant_23S_rRNA_mutation	4.41	21.57	42.44	0.23	31.34
A16S	Aminoglycosides	Aminoglycoside-resistant_16S_ribosomal_subunit_protein	1.56	5.9	47.11	0.13	45.3
ERMB	Macrolide-lincosamidestreptogramin	23S_rRNA_methyltransferases	0	4.39	91.06	0	4.55
RPOB	Rifampin	Rifampin-resistant_betasubunit_of_RNA_polymerase_RpoB	1.1	0.91	42.57	0.27	55.15
TUFAB	Elfamycins	EF-Tu_inhibition	1.68	2.79	53.84	0.07	41.63
O23S	Oxazolidinone	Oxazolidinone-resistant_23S_rRNA_mutation	7.1	22.31	43.5	0.55	26.54
TETM	Tetracyclines	Tetracycline_resistance_ribosomal_protection_proteins	0	0	100	0	0
GYRA	Fluoroquinolones	Fluoroquinolone-resistant_DNA_topoisomerases	1.06	0	29.42	0	69.51
SULI	Sulfonamides	Sulfonamide-resistant_dihydropteroate_synthases	0	0	96.58	0	3.42
APH3-DPRIME	Aminoglycosides	Aminoglycoside_O-phosphotransferases	0	5.36	84.64	0	9.99
TETX	Tetracyclines	Tetracycline_inactivation_enzymes	0	0	93.77	0	6.23
SULII	Sulfonamides	Sulfonamide-resistant_dihydropteroate_synthases	0	0.53	95.18	0	4.29
TETO	Tetracyclines	Tetracycline_resistance_ribosomal_protection_proteins	0	0	100	0	0
TETL	Tetracyclines	Tetracycline_resistance_MFS_efflux_pumps	0	9.89	90.11	0	0
ANT6	Aminoglycosides	Aminoglycoside_O-nucleotidyltransferases	0	4.15	86.55	0	9.3
TETQ	Tetracyclines	Tetracycline_resistance_ribosomal_protection_proteins	0	3.29	94.47	0	2.25
CMX	Phenicol	Phenicol_resistance_MFS_efflux_pumps	0	20	73.46	0	6.53
FLOR	Phenicol	Phenicol_resistance_MFS_efflux_pumps	0	0	55.47	0	44.53
PARE	Aminocoumarins	Aminocoumarin-resistant_DNA_topoisomerases	1.68	0	25.38	0	72.94
AAC6-PRIME	Aminoglycosides	Aminoglycoside_N-acetyltransferases	0	5.5	90.46	0	4.04
TETZ	Tetracyclines	Tetracycline_resistance_MFS_efflux_pumps	0	13.49	76.19	0	10.32
ANT4-PRIME	Aminoglycosides	Aminoglycoside_O-nucleotidyltransferases	0	20.56	78.14	0	1.3
FEXA	Drug_and_biocide_resistance	Drug_and_biocide_MFS_efflux_pumps	0	3.82	49.19	0	46.99
OPTRA	Multi-drug_resistance	Multi-drug_ABC_efflux_pumps	0	0	0	0	100
ERMF	Macrolide-lincosamidestreptogramin	23S_rRNA_methyltransferases	0	0	86.89	0	13.11
CFR	Macrolide-lincosamidestreptogramin	23S_rRNA_methyltransferases	0	8.91	91.09	0	0
ERMC	Macrolide-lincosamidestreptogramin	23S_rRNA_methyltransferases	0	6.93	93.07	0	0
MUXB	Multi-drug_resistance	Multi-drug_RND_efflux_pumps	0	0	35.38	0	64.62
APH6	Aminoglycosides	Aminoglycoside_O-phosphotransferases	0	2.5	82.47	0	15.03

Table 3. Correlation of mobile genetic elements (MGEs) with bacterial genera, antimicrobial resistance genes (ARGs), and virulence genes.

approaches to discern the functional relevance of detected resistance genes. Incorporating such analyses in future studies will provide a clearer understanding of the active resistome and its implications for antimicrobial resistance (AMR) risk assessment.

The presence of virulence factors poses a significant risk, as these genes can enhance bacterial pathogenicity, compounding the threat posed by antimicrobial resistance genes (ARGs)³⁹. The identification of virulence factors associated with immune modulation, effector delivery systems, and biofilm formation is particularly concerning⁴⁰. Immune modulation mechanisms can facilitate bacterial evasion of host immune responses, leading to persistent infections and increased transmission risks within farming environments⁴¹. The abundance of biofilm formation factors, which enhance bacterial adherence and resilience, indicates a potential for increased environmental persistence and resistance to disinfectants, further complicating disease control⁴². It is also evident that antimicrobial resistance is highly correlated with biofilm formation⁴³.

Significant differences in virulence factor abundance among samples are primarily attributed to the distinct environmental conditions and microbial dynamics within each sample type. For instance, sediment provides a stable, nutrient-rich environment conducive to the proliferation of bacteria with diverse virulence traits⁴⁴. This stability supports the maintenance and dissemination of both ARGs and virulence factors, explaining the higher abundance of immune modulation and effector delivery system factors in sediment⁴⁵. Conversely, the high prevalence of biofilm formation factors in feed and specific gastrointestinal samples can be linked to the constant microbial exposure and physical conditions that favor biofilm development⁴⁶. This enhances bacterial persistence and complicates control efforts⁴⁷.

The discovery of motility and adherence mechanisms, particularly in samples from droppings and chicken gut, suggests that bacteria in these environments are equipped to colonize and spread within the host and across environmental surfaces^{48,49}. These traits could lead to the dissemination of both pathogenic and resistant bacteria

within and beyond the farming system, representing a public health risk⁵⁰. Nutritional and metabolic factors, more abundant in sediment, highlight the role of environmental reservoirs in sustaining bacterial populations and serve as hotspots for the persistence and spread of both ARGs and virulence factors⁵¹.

The detection of exotoxin and invasion-related genes, although at lower levels, underscores the potential for harmful infections that could result from the spread of these virulence factors to humans or other animals⁵². The distinct distribution of virulence genes across different environmental compartments within the farming system reinforces the notion that environmental factors play a crucial role in the maintenance and dissemination of bacterial virulence and resistance⁵³.

In the context of the integrated chicken and fish farming system, the analysis of mobile genetic elements (MGEs) provides critical insights into microbial dynamics and mechanisms of horizontal gene transfer⁵⁴. The predominance of plasmids among the identified MGEs underscores their pivotal role in horizontal gene transfer, particularly in the dissemination of antimicrobial resistance genes⁵⁵. Their substantial presence within the system underscores their central function in propagating genetic traits across bacterial communities, thereby contributing to dissemination of genes including ARGs⁵⁶.

Insertion Sequences (IS) and Integrative and Conjugative Elements (ICE) are also evident within the system, albeit in lesser quantities relative to plasmids. IS elements are fundamental to the genome plasticity of microorganisms, driving genetic variability through insertional mutagenesis and subsequent genomic rearrangements⁵⁷. On the other hand, ICEs, which possess the capacity to integrate into host genomes and facilitate conjugative transfer, serve a specialized role in gene transfer⁵⁸. Despite their lower frequency, ICEs are integral to both vertical and horizontal gene dissemination processes within microbial communities⁵⁹.

Additionally, Transposons (Tn), Chromosomal Islands (CN), and Integrative Mobilizable Elements (IME), which further augment the diversity of MGEs. The distribution patterns of these MGEs across varied sample types reveal considerable environmental influence on their prevalence. The elevated abundance of MGEs in droppings and sediment suggests these environments act as prominent hotspots for gene exchange and microbial activity, likely attributable to elevated microbial biomass and metabolic activity⁶⁰. Conversely, the reduced MGE abundance in the fish intestine points to a diminished role in gene transfer within this specific environment. This variability in MGE distribution underscores the influence of environmental factors on microbial gene dynamics and elucidates the intricate interactions between microbial populations and their respective habitats⁶¹.

Transposons, such as Tn6072, Tn6246, and Tn4001, are critical vectors for the horizontal dissemination of antimicrobial resistance genes (ARGs)⁶². Their ability to transpose across different genomic locations facilitates the spread of resistance determinants such as *tetM*, *sull*, and *tetX*, thereby contributing to the persistence and proliferation of antimicrobial resistance within microbial populations⁶³. Integrative and conjugative elements (ICEs), including ICEBfrYCH46-1, play a crucial role in both vertical and horizontal gene transfer⁶⁴. Their capacity for genome integration and conjugative transfer highlights their importance in the exchange of resistance genes across bacterial communities, further reinforcing the genetic adaptability and survival of microbial populations in the farming environment. The role of transposons and plasmids extends to the transfer of virulence genes, with elements such as Tn125 and the plasmid *Nonomuraea gerezanensis* isolate nono1 plasmid I facilitating the dissemination of various virulence factors⁶⁵.

The association between specific microbial taxa and MGEs further elucidates the mechanisms of genetic exchange within microbial communities. Genera such as *Bacteroides*, *Clostridium*, and *Escherichia* exhibit significant interactions with MGEs, indicating their integral role in the propagation and distribution of these genetic elements⁶⁰. The distribution patterns of microbial phyla associated with MGEs reveal differential impacts on gene transfer processes across various environmental contexts. For instance, the predominance of Pseudomonadota in specific environments suggests a significant role in MGE-mediated genetic exchange⁶⁶, while the limited presence of other phyla, such as Mycoplasmatota and Campylobacterota, indicates a more restricted involvement in these processes⁶⁷.

This study provides a comprehensive reference genome for the chicken gut microbiome, offering critical insights into the diversity and functional potential of microbial communities within integrated farming systems. This resource serves as a foundation for comparative analyses across bacterial, archaeal, and viral domains, facilitating an in-depth exploration of the interplay between these microbial kingdoms within the gut environment. Such integrative analyses are essential for advancing our understanding of host-microbiota interactions and their roles in health and disease regulation in poultry systems⁶⁸.

The reference genome also holds significant potential for microbiome-based interventions, including the development of probiotics, precision feed additives, and targeted prebiotics to enhance poultry health and productivity^{69,70}. For example, probiotics specifically designed to align with the microbial composition of the chicken gut could effectively modulate the microbiota, reducing pathogenic colonization while improving nutrient absorption^{71,72}. Moreover, this genomic dataset enables the identification of auxiliary metabolic genes involved in the degradation of complex feed substrates, offering new avenues for optimizing feed formulations⁷².

Integrating this reference genome with multi-omics approaches could facilitate the identification of microbial markers for antimicrobial resistance (AMR) surveillance and management in farming systems⁷³. Such markers are instrumental in tracking the spread of resistance genes across interconnected environments, as highlighted in recent studies linking ARGs to specific microbial hosts and mobile genetic elements⁷⁴. Additionally, the inclusion of viral genomic data underscores the pivotal role of bacteriophages as vectors for horizontal gene transfer, thereby amplifying the dissemination of ARGs within the gut microbiome^{75,76}.

In contrast to earlier studies that predominantly explored the resistome diversity within chicken gut microbiomes, such as the work documenting the diversity of antibiotic resistance genes in live poultry markets and associated workers⁷⁷ or the comprehensive multi-kingdom microbiome catalog of the chicken gastrointestinal tract⁷⁸, this study broadens the scope by investigating the interplay of antimicrobial resistance

genes (ARGs), virulence factors (VFGs), and mobile genetic elements (MGEs) across interconnected terrestrial and aquatic environments.

Additionally, this study uniquely emphasizes the contribution of environmental reservoirs, such as sediment, to the persistence and spread of ARGs, highlighting their role as long-term sinks for resistance genes. Unlike previous works that focused on specific host-associated microbiomes^{79,80}, this study integrates cross-compartmental analysis, revealing how interconnected environments facilitate horizontal gene transfer via MGEs. Furthermore, the comprehensive assessment of VFGs alongside ARGs provides novel insights into the dual risk posed by pathogens capable of both resistance and heightened virulence, which has not been extensively explored in earlier studies.

By employing robust statistical analyses, such as the F-Welch and Games-Howell tests, this research also offers a quantitative evaluation of ARG distribution across sample types, strengthening the evidence for significant environmental differences in resistance gene abundance. This multi-dimensional approach contributes to a deeper understanding of how integrated farming practices shape microbial community dynamics and resistance profiles, setting this study apart from previous investigations.

In summary, the role of MGEs in the dissemination of ARGs and virulence genes is integral to understanding microbial adaptability and the dynamics of genetic exchange within integrated farming systems. The findings highlight the essential contribution of MGEs to microbial gene flow and the ecological complexity of microbial communities.

Conclusion

In conclusion, this reveals a concerning prevalence of antimicrobial resistance genes (ARGs) and virulence factors throughout the farming system, with MGEs playing a pivotal role in their dissemination. Poultry droppings emerge as the most significant source of both ARGs and virulence factors, underscoring the critical need for meticulous management of waste. In contrast, fish intestines and feed contribute minimally to the spread of resistance genes. Therefore, stringent antibiotic stewardship and careful handling of poultry droppings are essential measures recommended by this study.

Methodology

Ethical approval

The study was approved by the Ethical Review Committee of the Faculty of Biological Sciences at Jashore University of Science and Technology, Jashore, Bangladesh (certification number: ERC/FBST/JUST/2024–200). All experimental protocols were reviewed and approved by this committee. All methods were carried out in accordance with the ARRIVE (Animal Research: Reporting of In Vivo Experiments) and AVMA (American Veterinary Medical Association) guidelines. All possible measures were taken to minimize pain and harm to the experimental animals. All experimental methods were carried out in accordance with relevant guidelines and regulations.

Sample collection and total DNA extraction

Three integrated fish farms incorporating chicken farming, located in Solua Bazar, Jhikargacha, and Khajura within the Jashore district of Bangladesh, were selected for this study, designated as Farm-1, Farm-2, and Farm-3, respectively. Five distinct sample types were collected from each farm, including feed ($n=1$), chicken gut ($n=2$), chicken droppings ($n=2$), fish intestine ($n=2$), and pond sediment ($n=2$), resulting in nine samples per farm and 27 samples in total. Feed, chicken droppings, and sediment were aseptically collected using sterile containers to prevent contamination. Chicken droppings were sampled immediately post-defecation using sterile spatulas. Chicken euthanasia was performed following the AVMA Guidelines for the Euthanasia of Animals (2020 Edition). Birds were placed in a CO₂ chamber prefilled to 30% of its volume, with CO₂ concentration increased at a rate of 30–70% of chamber volume per minute. Death was confirmed through the cessation of respiratory movements and absence of the corneal reflex. Dissection was performed using sterile instruments to obtain gut sections.

Fish euthanasia also adhered to AVMA guidelines, using a CO₂-enriched tank with gradual gas diffusion to ensure sedation and euthanasia. Fishes were selected and acclimated in laboratory tanks maintained at 26 ± 2 °C, with pH 7.0–7.5 and dissolved oxygen levels above 6 mg/L. Observations included loss of equilibrium, cessation of opercular movement, and secondary decapitation for confirmation of death. Intestine samples were dissected using sterile instruments. Pond sediment samples were collected from multiple pond locations using sterile scoops. All samples were stored at 4 °C immediately after collection and transported to the laboratory within 1–2 h. For downstream analysis, samples of the same type were pooled within each farm, resulting in five composite samples per farm. DNA extraction was performed using the DNeasy PowerSoil Pro Kit and QIAamp Fast DNA Stool Mini Kit (QIAGEN) per manufacturer protocols. Extracted DNA was stored at -20 °C for sequencing. Detailed sampling site and sample descriptions are available in Supplementary Table 1.

Library Preparation and shotgun metagenomic sequencing

Shotgun metagenomic libraries were prepared using the iNextEra library preparation protocol. Briefly, the DNA was tagged with Illumina bead-linked transposomes (Illumina DNA library prep, Illumina, Inc., San Diego, CA, USA) and 2X TMP buffer⁸¹ in a total volume of 6 µL (3 µL of 2X TMP buffer, 0.5 µL of Bead-linked transposome and 2.5 µL of DNA) and incubation at 53 °C for 30 min. Following tagmentation, a limited-cycle PCR step completed the Illumina-compatible adapters with dual-indexes using custom iNextEra primers for sequencing of pooled libraries. Each 51.5 µL of PCR master mix included 40.25 µL HyClone water (Cytiva Life Sciences, Marlborough, MA, USA), 9.5 µL of Q5 Reaction Buffer (New England Biolabs, Ipswich, MA, USA),

1.25 μL of 10mM dNTPs (Roche Diagnostic Corporation, Indianapolis, IN, USA), and 0.5 μL of Q5 High-Fidelity DNA Polymerase (New England Biolabs, Ipswich, MA, USA). The master mix was then added to 6 μL of tagmentation mix. Finally, 2.5 μL of 5 μM iNextEra5 index primer and 2.5 μL of 5 μM iNextEra7 index primer were added to the PCR mix, creating unique index combinations in a total PCR volume of 62.5 μL . The limited-cycle PCR was done using a T100 thermal cycler (Bio-Rad Laboratories, Inc., Hercules, CA, USA) with an initial strand repair at 72 $^{\circ}\text{C}$ for 3 min, denaturation at 98 $^{\circ}\text{C}$ for 3 min followed by 12 cycles consisting of 45 s at 98 $^{\circ}\text{C}$, 30 s at 62 $^{\circ}\text{C}$, and 2 min at 72 $^{\circ}\text{C}$, concluding with a final extension at 72 $^{\circ}\text{C}$ for 1 min. After PCR, 5 μL of PCR product was run on an agarose gel to ensure each sample amplified, yielding a smear in the desired size-range (300–700 bp). Then, the libraries were quantified using Qubit Flex (Thermo Fisher Scientific, Waltham, MA USA), normalized, pooled and purified using AMPure XP beads (Beckman Coulter, Inc., Brea, CA, USA). The final pool was quantified and sent to Novogene (Novogene Corporation Inc., Sacramento, CA, USA) to obtain 2×150 bp paired-end reads using Illumina NovaSeq X plus sequencing system (Illumina, Inc., San Diego, CA, USA).

Quality control and identification of MGEs, ARGs, VFGs and bacterial taxa

The generated FASTQ files were evaluated for quality using FastQC v0.11.6⁸². Adapter sequences, and low-quality ends per read were trimmed by using Trimmomatic v0.39⁸³ with a sliding window size of 4; a minimum average quality score of 20; minimum read length of 40 bp (base pairs). After quality control, there were an average of 129.63 million pairs of reads (Minimum = 89.15 million and Maximum = 204.47 million reads). The abundance of mobile genetic elements (MGEs), antimicrobial resistance genes (ARGs), and virulence factors were identified by AMR++ bioinformatic pipeline (<https://github.com/Microbial-Ecology-Group/AMRplusplus>)⁸⁴. In this pipeline, the Burrows-Wheeler Aligner (BWA) was used to align sequence reads to host genomes, such as the chicken (*Gallus gallus*) GRCg6 and the common carp (*Cyprinus carpio*) GCF_018340385.1, in order to filter out these reads. Then the reads were aligned with the MobileElementFinder MGEdb database (v1.0.2)⁸⁵, the MEGARes database (v2.0)⁸⁴, the updated Virulence Factor Database (VFDB)⁸⁶ as of August 2024, plasmid detection PLSDB (v. 2023_11_03_v2)⁸⁷. Only genes that met a stringent identification threshold of 90%—defined as the percentage of genes matched by at least one sequencing read—were retained for analysis using the Resistome Analyzer (<https://github.com/cdean/resistomeanalyzer>). Bacterial taxonomic classification was conducted using CZID⁸⁸ formerly known as IDseq, where only sequences with an NT% identity above 90% and alignment length (NRL) greater than 50 were considered. This classification process was further refined using Taxonkit⁸⁹ for accurate taxonomic resolution.

Statistical analysis and data visualization

Statistical analyses and visualizations were performed using R Studio version 4.3.3⁹⁰. To ensure the normality of our data, we applied the Total Sum Scaling (TSS) normalization method. The TSS normalization equation is expressed as follows:

$$X_{ij}^{TSS} = \frac{X_{ij}}{\sum_{j=1}^n X_{ij}}$$

Where:

X_{ij} : The raw count or value of feature j in sample i . $\sum_{j=1}^n X_{ij}$: The sum of all feature values in sample i .

X_{ij}^{TSS} : The normalized value of feature j in sample i .

A range of plots was employed to analyze and present the data comprehensively. A chord diagram was constructed using the “circlize”⁹¹ package, with the “RColorBrewer”⁹² package providing enhanced color schemes. For visualizing distributions and conducting statistical comparisons, violin plots were created with the “ggstatsplot”⁹³ package. This package allowed for the integration of statistical tests, including Welch’s test⁶⁸, Kruskal-Wallis test⁹⁴, Games-Howell test⁹⁵, Dunn⁹⁶ and Bayesian analysis⁹⁷. Welch’s test was utilized to compare means between two groups assuming unequal variances, while the Games-Howell test was used for pairwise comparisons in cases of unequal variances and sample sizes. Bayesian analysis provided a probabilistic approach to inferential statistics, incorporating prior knowledge and estimating the credibility of hypotheses. Heatmaps were generated using the “pheatmap”⁹⁸ package, with the “RColorBrewer” package contributing color schemes and clustering functionalities. Correlation heatmaps, depicting Spearman correlations⁹⁹, were plotted with the “ggcorrplot”¹⁰⁰ package, supported by the “ggplot2”¹⁰¹ package for enhanced visualization. Spearman correlation analysis was employed to assess monotonic relationships between variables, providing a non-parametric measure of association. The relative abundance plot was created using the “ggplot2” package, which facilitated the graphical representation of the proportion of different components within the dataset.

Limitations and future prospects

This study’s reliance on metagenomic analysis offers a broad genetic overview but lacks insight into the functional activity of identified genes. While the detection of antimicrobial resistance genes (ARGs) and virulence factors (VFGs) provides valuable information, the inability to validate their expression or phenotypic implications limits our understanding of their real-world impact. The research is geographically limited, focusing solely on integrated farming systems in Bangladesh, which may constrain the broader applicability of its findings to other regions with differing practices or environmental conditions. Additionally, the cross-sectional design captures only a single time point, potentially overlooking temporal variations in gene abundance and dynamics across seasons or farming cycles. Future studies should aim to integrate metagenomic data with transcriptomic, proteomic, or metabolomic analyses to link gene presence with functional activity and phenotypic outcomes.

Expanding the scope to diverse geographical regions and varying integrated farming practices will provide a more comprehensive understanding of ARG and VFG dissemination. Additionally, adopting longitudinal study designs will help elucidate the temporal dynamics of these genes, offering deeper insights into their persistence and spread. Exploring the role of environmental co-factors, such as heavy metals and biocides, alongside ARGs, could also shed light on additional drivers of resistance proliferation.

Data availability

The data that support the findings of this study are openly available in NCBI BioProject at <https://www.ncbi.nlm.nih.gov/bioproject/> reference number PRJNA1165771.

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Author contributions

SMK carried out the bioinformatics analysis, generated figures, curated data, and drafted the original manuscript. KTI performed the analysis of ARGs and MGEs, formatted taxonomic data, and contributed to manuscript writing. RNR analyzed virulence genes, prepared data for plotting, and contributed to manuscript writing. MIUB performed shotgun sequencing and contributed to data analysis, manuscript review, and editing. TC participated in data analysis, manuscript reviewing, and editing. MSR OKI and MTI developed the research concept, finalized the bioinformatics pipelines, interpreted the results, and contributed to manuscript review and editing. MTI conceived the research, secured reagent support, thoroughly reviewed the manuscript, and led the overall project.

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Declarations

Competing interests

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT 4.0/Paraphrasing service in order to improve writing quality. After using this service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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